

# STORYBOARD — Mapping Influence for Marketing Growth

**Prepared for:** Ana Sofia Mendoza — Specialist in Digital & Performance Marketing

## Executive Summary

Ana Sofia Mendoza, a Digital & Performance Marketing Specialist, is seeking to strengthen her data-driven decision-making process by leveraging network analytics to optimize influencer selection, campaign targeting, and growth strategies.

This storyboard presents four key visualizations derived from a Facebook network dataset, each designed to answer two fundamental business questions:

1. Which individuals hold the highest potential to amplify reach and credibility across digital audiences?
2. How do different types of network influence (visibility, connectivity, and strategic position) impact campaign performance and ROI?

Each visualization builds upon the previous one, progressing from a strategic overview of network structure to tactical influencer targeting and campaign execution.

Together, they form a coherent analytical framework that enables Ana Sofia to move from insight to action: identifying high-value partners, segmenting communities, and aligning marketing investments with measurable outcomes in performance marketing.

## Big Idea

Strategic user selection for marketing campaigns must prioritize individuals with high bridging capacity across communities, as their network position directly enhances reach, influence, and campaign effectiveness.

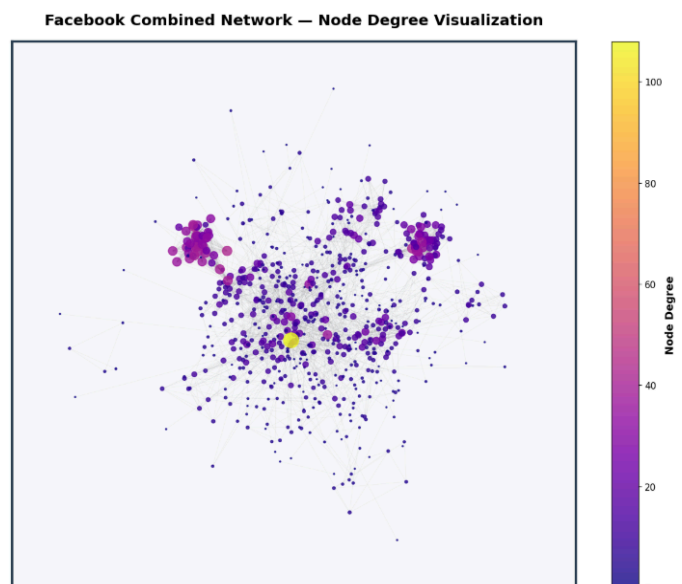
## 1. Network Landscape Overview

### Visualization: "Facebook Combined Network — Node Degree Visualization"

**Business Question:** What is the fundamental structure of our network, and how efficient is overall connectivity for campaign diffusion?

What the Visualization Shows:

This is the initial shot, a bird's-eye view of the social network (a sample of 800 nodes). The size and color of each node are scaled directly by its



Degree Centrality (the number of direct connections, i.e., local popularity) . Faint lines (edges) illustrate the vast interconnections. This scene establishes the network's context.

### Key Insights:

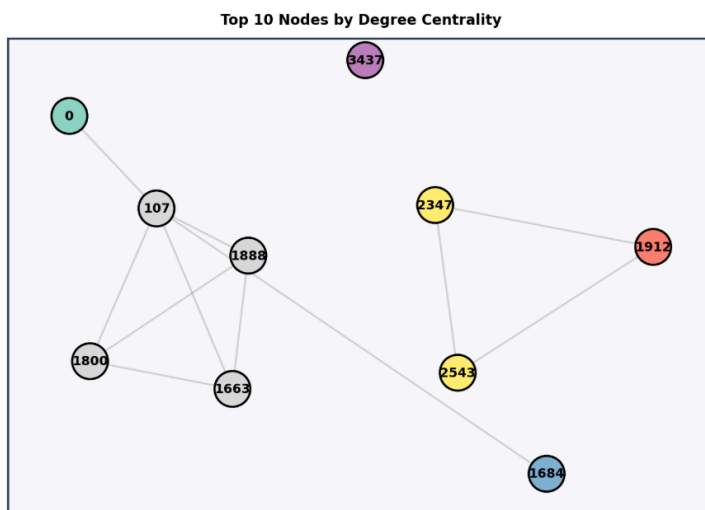
- **'Small-World' Structure:** The network exhibits "small-world" characteristics, meaning the average path length between any two users is short. This suggests that campaigns have the potential to **diffuse quickly** if launched from the correct point.
- **Concentration of Popularity (Pareto Principle):** We observe a clear concentration of influence: **very few nodes** (large and bright) hold the vast majority of the connections. The overall density is low (aprox 0.0108), confirming that the reach of any campaign will depend critically on successfully identifying these few social *hubs*.
- **Foundation for Targeting:** Campaign *reach* is primarily held by this small elite.

Decision Point:

Validate the "Hub & Spoke" Strategy (Concentration on Hubs): Should we allocate the majority of the influencer marketing budget (> 80%) to this small percentage of the largest, most connected nodes to ensure maximum visibility?

## 2. Leadership and Community Structure

Visualization: "Top 10 Nodes by Degree Centrality"



**Business Question:** How is power distributed in the network, and what is the role of these "social bridges" in information diffusion?

What the Visualization Shows:

This shot focuses on the "who." It isolates the Top 10 nodes by Degree Centrality (the most popular). Color now represents the Community ID of each node (detected by the Louvain algorithm) . This helps differentiate local popularity from structural influence.

### Key Insights:

- **Multiple Power Centers:** Node 107 stands out as the primary leader (highest Degree), but not all top nodes belong to the same community (different colors). This confirms the existence of **multiple power centers** that must be targeted.

- **Role of Bridges:** Several top nodes are clearly intersections between communities (with links crossing color boundaries). For example, Node 1684 acts visibly as a structural bridge, making it crucial for viral diffusion outside the primary group.
- **High Modularity ( $Q = 0.8350$ ):** The exceptionally high modularity value confirms that ties within groups (same color) are much denser than ties between them. This indicates that campaigns could stagnate if they fail to leverage the correct bridge-influencers.

Decision Point:

Define Strategic Influencer Profiles: Identify and prioritize nodes with high degree that also function as bridges between the 16 detected communities. These are the Strategic Influencers (combining local popularity with global diffusion capability).

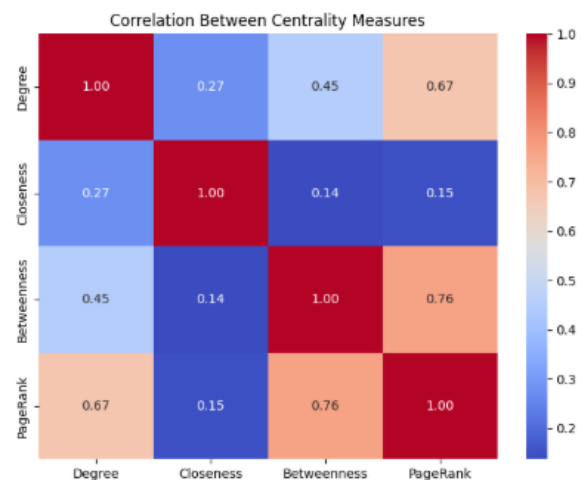
### 3. The ROI Model for Influence

**Visualization: "Correlation Between Centrality Measures"**

**Business Question:** Which metric of influence best aligns with the goal of *Performance Marketing* (driving action and strategic reach)?

What the Visualization Shows:

This shot provides quantitative justification. A Heatmap visualizes the correlation matrix between the four key centrality metrics (Degree, Closeness, Betweenness, and PageRank).



**Key Insights:**

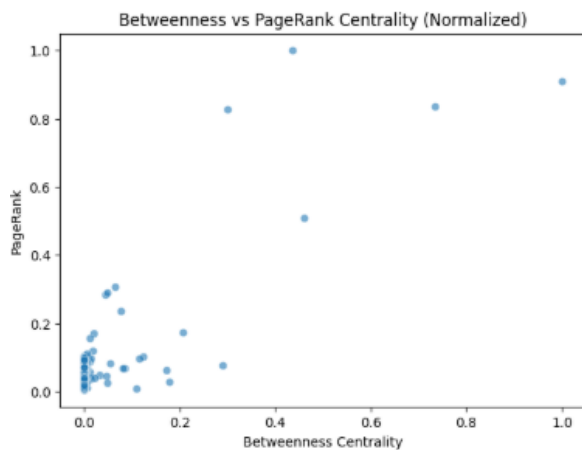
- **Betweenness ↔ PageRank (0.76):** This is the strongest correlation and the key insight. Nodes that act as bridges (high Betweenness) are also those with the greatest authority in information flow (high PageRank). This defines the ideal influencer profile for a viral diffusion campaign.
- **Degree ↔ PageRank (0.67):** The moderate correlation indicates that being popular is not enough; the quality of connections (PageRank) is a more critical factor than just the quantity (Degree).
- **Closeness ↔ Betweenness (0.14):** Very weak correlation. Nodes physically close to others in the network (high closeness) are not necessarily the ones that control the flow of information (high betweenness).

Decision Point:

Adjust the Influence Model: Our selection model must heavily weigh PageRank and Betweenness over Degree or Closeness. These metrics identify the "social glue" that ensures maximum diffusion among communities with the necessary authority to drive conversion.

## 4. Budget Allocation Dashboard

### Visualization: "Betweenness vs PageRank Centrality (Normalized)"



**Business Question:** How should we segment our *influencers* to allocate budgets accurately and maximize campaign effectiveness?

What the Visualization Shows:

This is the final shot and the tactical action dashboard. It is a scatter plot of Betweenness vs. PageRank (both normalized to the maximum) .

#### Key Insights:

- **The Elite (Top Right Quadrant):** Nodes in this area (near 1.0 on both axes) are the "Super-Connectors." They possess maximum authority and bridging capacity. They are a limited resource but guarantee exceptional ROI in diffusion.
- **Silent Bridges (High Betweenness, Low PageRank):** These nodes connect groups but lack high authority. They are useful for low-cost reach or content seeding campaigns.
- **Risk Mitigation:** The highest Betweenness nodes are structural bridges; relying solely on them creates a vulnerability (if they are removed, the network fragments). Diversification is necessary.

Decision Point:

Campaign Strategy and Budget Allocation:

1. **Premium Segment (High PageRank / High Betweenness):** Allocate the largest budget for conversion and brand lift campaigns, leveraging their authority and strategic reach.
2. **Reach Segment (High Betweenness / Low PageRank):** Use for reach and awareness campaigns as a low-cost complement to ensure coverage across all communities.

### General Key Insights from Network Analysis

The analysis of the Facebook network reveals that influence is not solely based on popularity, but on **strategic position** and **authority** within a highly modular structure.

- **Influence is Not Just Popularity (The PageRank Filter):** The correlation between Degree Centrality (popularity/follower count) and PageRank (authority/influence) is only moderate (0.67). This indicates that many popular users lack the necessary network authority to drive widespread conversion or diffusion.

- **The Optimal Influencer is a Strategic Bridge:** The strongest correlation is between Betweenness Centrality (bridging capacity) and PageRank (0.76). The most effective influencers are those who act as bridges between different communities and possess high authority.
- **High Network Segregation (Modularity):** The network is organized into 16 distinct, tightly-knit communities with a high modularity score (0.8350). This means information struggles to flow organically between groups, making bridge nodes essential for cross-community reach.
- **Clicquish Local Dynamics (Deep Trust):** The average Clustering Coefficient is high (0.6055), with a large number of nodes in perfect cliques ( $C(v)=1.0$  peak). This strong local structure requires a "Deep Trust" messaging strategy focused on social validation and personal recommendation to penetrate these circles.
- **Vulnerability of Centrality:** The network's global efficiency rests on a very small number of Super-Connectors, whose loss or underperformance poses a significant risk of network fragmentation.

## Strategic Roadmap for Performance Growth

### Phase 1: Tactical Influencer Selection (Optimization)

The initial phase focuses on optimizing partner selection by shifting away from the traditional metric of follower count. The first crucial action is to build a Strategic Influencer Index where partners are prioritized based on a weighted index: PageRank (Authority) and Betweenness (Bridging) will carry more weight than basic Degree (Popularity). The success of this selection will be measured by Index Score Benchmarks, ensuring all new partners exceed specific thresholds (e.g., Target all candidates with PR > 0.8 and Betweenness > 0.6).

Following selection, we must diversify bridge allocation. Given the network's segregation, we will distribute the reach/awareness budget across a minimum of 5-7 bridge-influencers to mitigate the risk of relying on a single structural node. This diversification will be monitored by the Risk Mitigation Score, which tracks campaign coverage across the 16 identified communities. Finally, we must implement a Tiered Budget Assignment strategy, where the highest budget goes to Tier 1 (Premium) "Super-Connectors" (High PageRank / High Betweenness) to drive conversion, while Tier 2 (Reach) partners (high Betweenness / low PageRank) are utilized for low-cost awareness. The performance will be evaluated by the Conversion Rate (CR) from these tiered partners.

### Phase 2: Messaging and Monitoring (Execution)

The second phase concentrates on the campaign execution and long-term success metrics. The primary execution step is to implement a "Deep Trust" Messaging strategy. This involves developing creative concepts that rely heavily on personal testimonial, social proof, and high perceived relevance, aligning the message with the observed high local clustering (cliques). The effectiveness of this approach will be measured by the Engagement Rate (ER) within the influencers' local community.

To validate the entire strategy, we will launch a Pilot Campaign & Benchmark using the newly developed Strategic Influencer Index. The main success metric here is the Cost Per Acquisition (CPA) achieved via the SII-selected partners. For long-term risk assessment, we must monitor Network Change. This involves

tracking the Modularity Change Rate over time; a successful campaign should show a slight decrease in modularity, indicating that the bridge-influencers are effectively creating new links between segregated communities. This resilience will be assessed via the Modularity Score and Network Density 6 months post-campaign.